ORGANIC CHEMISTRY
ASYNCHRONOUS ACTIVITY 2

One Molecule, Four Representations:
From Molecular Formula to Bond-Line Structure

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BSChE 2A



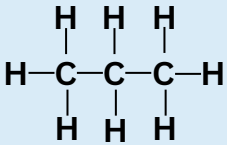
CARD 01 — PROPANE

Molecular Formula: C_3H_8

1 MOLECULAR FORMULA

C_3H_8

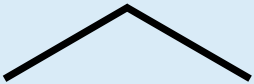
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

4 BOND-LINE FORMULA



✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed

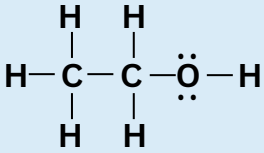
CARD 02 — ETHANOL

Molecular Formula: C_2H_6O

1 MOLECULAR FORMULA

$$C_2H_6O$$

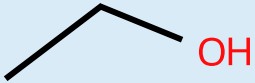
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2OH$$

4 BOND-LINE FORMULA



- Atom Count Verified
- Valence Checked
- Bond order checked
- Same connectivity confirmed

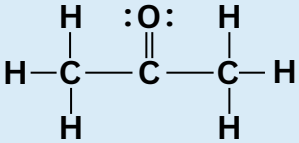
CARD 03 — ACETONE (PROPAN-2-ONE)

Molecular Formula: C_3H_6O

1 MOLECULAR FORMULA

$$C_3H_6O$$

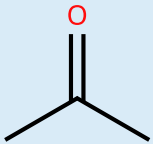
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3COCH_3$$

4 BOND-LINE FORMULA



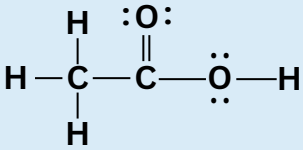
✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed

CARD 04 — ACETIC ACID (ETHANOIC ACID) *Molecular Formula: $C_2H_4O_2$*

1 MOLECULAR FORMULA

$$C_2H_4O_2$$

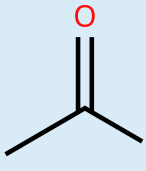
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3COOH$$

4 BOND-LINE FORMULA



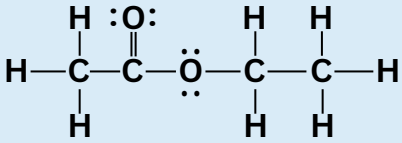
✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed

CARD 05 — ETHYL ACETATE (ETHYL ETHANOATE) *Molecular Formula: $C_4H_8O_2$*

1 MOLECULAR FORMULA

$$C_4H_8O_2$$

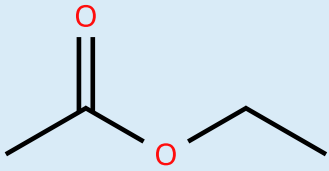
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3COOCH_2CH_3$$

4 BOND-LINE FORMULA



✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed

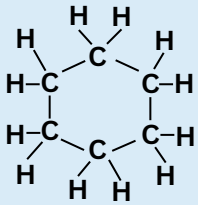
CARD 06 — CYCLOHEXANE

Molecular Formula: C_6H_{12}

1 MOLECULAR FORMULA

$$C_6H_{12}$$

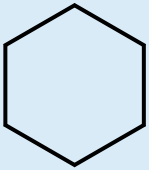
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$(CH_2)_6$$

4 BOND-LINE FORMULA



✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed

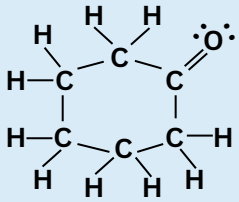
CARD 07 — CYCLOHEXANONE

Molecular Formula: $C_6H_{10}O$

1 MOLECULAR FORMULA

$$C_6H_{10}O$$

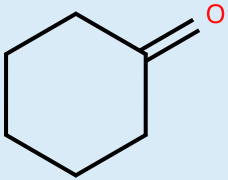
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$(CH_2)_5CO$$

4 BOND-LINE FORMULA



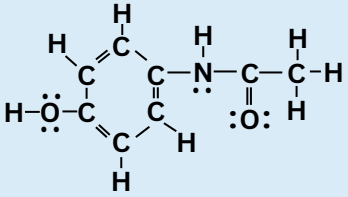
✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed

CARD 08 — ACETAMINOPHEN (PARACETAMOL) *Molecular Formula: $C_8H_9NO_2$*

1 MOLECULAR FORMULA

$$C_8H_9NO_2$$

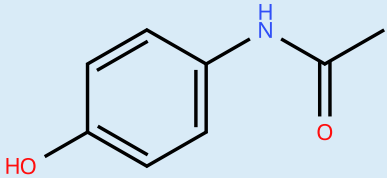
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$HOC_6H_4NHCOCH_3$$

4 BOND-LINE FORMULA



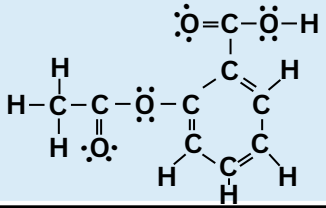
✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed

CARD 09 — ASPIRIN (ACETYLSALICYLIC ACID) *Molecular Formula: $C_9H_8O_4$*

1 MOLECULAR FORMULA

$$C_9H_8O_4$$

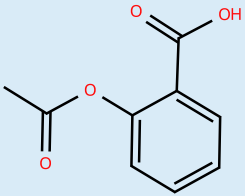
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$CH_3COOC_6H_4COOH$$

4 BOND-LINE FORMULA



✓ Atom Count Verified ✓ Valence Checked ✓ Bond order checked ✓ Same connectivity confirmed

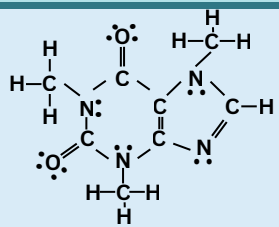
CARD 10 — CAFFEINE

Molecular Formula: $C_8H_{10}N_4O_2$

1 MOLECULAR FORMULA

$$C_8H_{10}N_4O_2$$

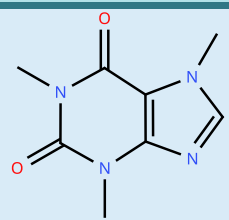
2 LEWIS STRUCTURE



3 CONDENSED STRUCTURAL FORMULA

$$C_8H_{10}N_4O_2 \text{ or } C_5HN_4O_2(CH_3)_3$$

4 BOND-LINE FORMULA



- ✓ Atom Count Verified
- ✓ Valence Checked
- ✓ Bond order checked
- ✓ Same connectivity confirmed